

# Clarifying RNA targets for antisense design in extraskeletal myxoid chondrosarcoma

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To the Editor,

Antisense oligonucleotides (ASOs) are short molecules designed to bind a chosen RNA sequence. A cancer fusion creates a junction between RNA sequences from two genes. An ASO spanning that junction is a possible way to target the fusion. Before testing such a design, however, we need to know the exact joined sequence and how closely it resembles normal RNA. We investigated these questions in extraskeletal myxoid chondrosarcoma (EMC).

We revisited 35 designs from a catalogue built mainly from hypothetical junctions. We selected junctions that could be linked to deposited fusion sequences or published model descriptions. Some sequences were directly recorded. Others had to be reconstructed from reference coordinates or a description of an engineered construct. We kept these kinds of evidence separate and examined several ASO binding positions across each junction.

The first finding concerned the published USZ20-EMC1 cell model.[1] Its fusion was described with exon numbers, which identify segments of the RNA. These numbers depend on the reference transcript used. After reconciling the references, we found that the reported junction corresponded to an existing catalogue design. Simply copying the exon numbers between references would instead produce a different target sequence. In one comparison, the resulting ASOs differed at 11 of their 16 bases.

The published figure therefore contains more information than we had previously recognized. Its coordinates and sequence identifiers allow a reference junction to be reconstructed. This requires assumptions about the genome reference and how the coordinates mark the junction. It does not establish the patient's RNA sequence, which could contain variants or extra inserted bases. The exact junction in the companion model, USZ22-EMC2, could not be resolved from its figure.

The second finding concerned normal RNA. The original comparison used one transcript per parent gene. We expanded it to include alternative RNA versions, known as isoforms. Each 16-base ASO design contains a central six-base DNA region, called the gap. We compared its intended RNA target with nonfusion reference RNAs. We measured the longest stretch of consecutive matching bases that covered the entire gap.

With additional isoforms included, this match became longer for 9 of 35 designs. For one TCF12–NR4A3 design, it increased from seven to eleven bases. The longer match came from an alternative NR4A3 transcript described in earlier work.[2] The five USZ20 comparisons were unchanged. No design had a complete 16-base match in the expanded set.

The choice of reference RNA therefore affects how distinct a fusion target appears from normal RNA. Checking for unintended RNA matches is already standard practice.[3] This audit identifies a specific model correspondence and the normal RNA versions that change comparisons for existing designs. It does not update the whole catalogue or estimate patient coverage. The supplement and public archive provide the exact sequences, sources and analysis code.[4]

We compared reference RNA from the fusion-partner genes only. We did not screen all human transcripts or measure expression in healthy tissue. No ASO was synthesized or tested. These sequence results cannot establish whether an ASO will cleave its target, spare normal RNA or be safe.

### **Declarations**

The author directed this public-data study. It involved no new recruitment or experiments and received no external funding. The author has no competing financial interests and declares a nonfinancial interest as an EMC survivor. Claude assisted the preceding resource. OpenAI Codex with GPT-6-Astra assisted retrieval, analysis, verification and drafting. Computational checks and an independent AI critique do not constitute experimental validation or human peer review. The author is responsible for the manuscript.

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